Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. What is a potential consequence of simple imputation strategies like replacing missing values with zero? 1 / 1 point

- (•) Altering the data distribution
- ( ) Enhancing model performance
- ( ) Making the dataset larger
- ( ) Reducing computational complexity

Nice work

Simple imputation strategies can significantly change the distribution and interaction characteristics of the variables in the dataset, introducing biased noise and affecting the covariance structure.

2. What strategy would allow you to utilize information from multiple columns in a dataset to impute missing values? 1 / 1 point

- (•) Multivariate approach
- ( ) Listwise deletion of multiple incomplete rows
- ( ) Univariate approach
- ( ) Single column replacement

Nice work

A multivariate approach involves looking across multiple columns to infer missing values in a target column, which is suitable for handling missing data in a multivariate dataset.

3. What does the term 'Missing Completely at Random' (MCAR) imply about the missing data? 1 / 1 point

- ( ) The missingness occurs in a pattern related to one or more variables.
- ( ) The missingness is related to specific values or ranges of the missing data.
- (•) The missingness is unrelated to any specific variable or feature.
- ( ) The missingness depends on the observed data only.

Nice work

Missing Completely at Random (MCAR) means that the missing data points in a dataset are not related to any specific variable or feature: they are an entirely random subset of the entire dataset.

4. How does the 'Estimator' property enhance the functionality of IterativeImputer? 1 / 1 point

- ( ) Enhances the accuracy of imputation
- ( ) Improves computational speed
- (•) Allows customization of the regression method

Nice work

The 'Estimator' property lets users specify a custom regression method, such as a random forests regressor, to be used in the imputation process, providing flexibility and potentially improving imputation accuracy.

5. A team is developing a model to predict house prices, but some of the variables have missing data. They decide to use a pipeline that includes an imputer followed by a regressor. What is the primary benefit of using a pipeline in this scenario? 1 / 1 point

- (•) Seamless integration of imputation and regression
- ( ) Increased computational efficiency
- ( ) The primary benefit of integrating an imputer and a regressor in a pipeline is to ensure that the imputation of missing values and the regression analysis are handled in a cohesive manner, which helps in maintaining the integrity of the model's training process. It is not inherently guaranteed to enhance the accuracy of the model; model accuracy depends on various other factors including the choice of imputer, regressor, and the data itself.
- ( ) Automatic feature selection

Nice work

Using a pipeline that combines an imputer and a regressor allows for seamless data processing where missing values are filled before regression analysis, simplifying the workflow and ensuring data integrity.

6. When using the IterativeImputer in Scikit-learn, what is the initial step in the imputation process? 1 / 1 point

- ( ) Normalize all feature columns
- ( ) Calculate the mean of all features
- ( ) Remove rows with missing values
- (•) Select a feature column as the output y

Nice work

The first step in using IterativeImputer is to pick a feature column to act as the output y, which will be the target for which the missing values are predicted.

7. What is an effective approach to imputing missing data when human inspection can identify regularities in error patterns and/or clear relationships between columns in a dataset? 1 / 1 point

- ( ) Statistical modeling
- ( ) Complete case analysis
- ( ) Machine learning-based imputation
- (•) Data wrangling tools

Nice work

Data wrangling tools allow for scripting to automate the process of filling in missing values based on detectable regularities between columns, which is effective when replacements for missing values are obvious

8. If you decide to move ahead with a complete case analysis on a dataset with missing values, what is an important factor to consider before applying this method? 1 / 1 point

- ( ) Category of missing data mechanism (e.g. missing at random or not)
- ( ) Number of variables in the dataset
- ( ) Presence of outliers in the data
- (•) Proportion of missing values

Nice work

Before applying complete case analysis, it is crucial to consider the proportion of all rows that have missing values to ensure that you do not lose a significant amount of data.

9. If a dataset contains pairs of data points and the goal is to optimize an objective function that involves using the dissimilarity between these points, what should be done if there is missing data? 1 / 1 point

- (•) Filter out pairs that have at least one data point with missing data (by setting the pair's weight to zero in the objective function)
- ( ) Remove all data points with missing data from the dataset (by setting the data point's weight to zero in the objective function) before computing pairwise dissimilarity.

Nice work

If there is missing data that prevents the computation of dissimilarity between two points, the weight w(i, j) in the objective function assigned to that pair should be set to zero, effectively filtering out those pairs from contributing to the objective.